FVS-ACD user's guide



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About

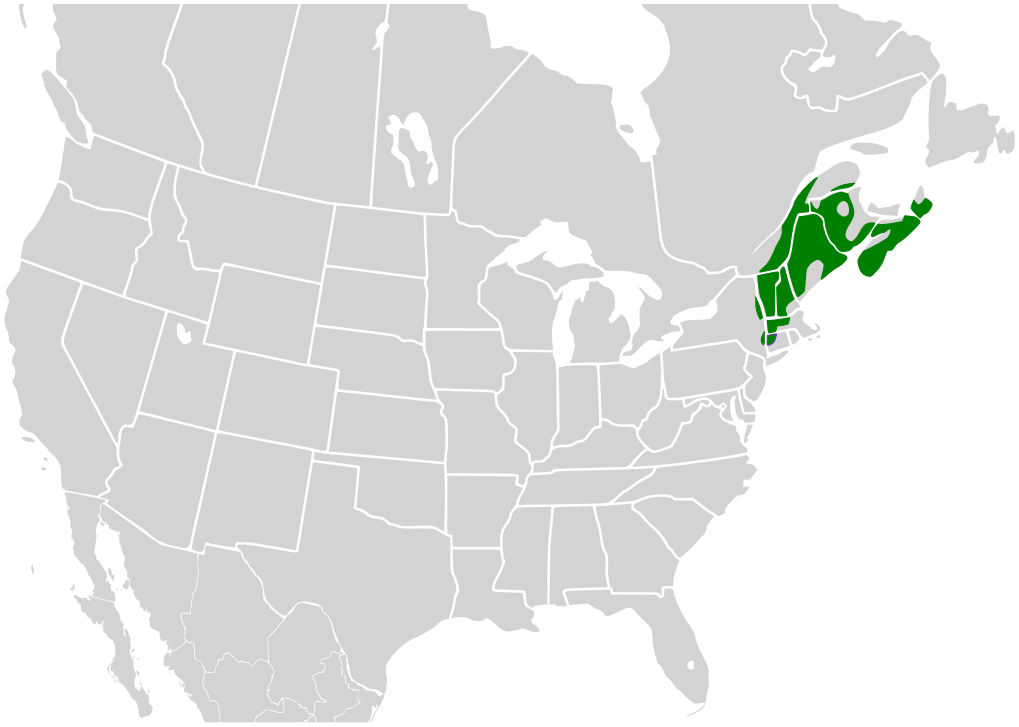


Figure 1. Acadian forest region
commons.wikimedia.org/wiki/File:New_England-Acadian_forests_map.svg

The Acadian forest region comprises northern New England and Atlantic Canada and is a transitional mixed conifer and hardwood forest type located between northern hardwood forests to the south and west and boreal coniferous forest to the north (Loo and Ives 2003). This mix of forest types and a long and varied management history across the region creates a complex setting for growth projections.

Researchers and practitioners have long appreciated the need for a growth and yield model specific to the Acadian region. Fortunately, members of the University of Maine's Cooperative Forestry Research Unit (CFRU) have supported the development of the Acadian model. A team of researchers and graduate students led by Aaron Weiskittel has developed a growth and yield model for the region. The model also addresses regional growing conditions, including modifiers for commercial thinning, spruce budworm defoliation, and hardwood form, and risk. The Acadian model now represents the best available science for the projection of forest stand conditions across the Acadian region.

This user's guide is intended to provide a brief overview of the Acadian model and its use within the Forest Vegetation Simulator (FVS) framework. It assumes minimal experience with FVS.

Forest Vegetation Simulator (FVS)

FVS is a publicly available growth model, developed and maintained by the US Department of Agriculture, Forest Service. FVS is a semi-distance-independent individual tree growth and yield model. This means individual trees are projected forward in time, rather than aggregated plot or stand data.

Individual tree models allow much flexibility in projecting a wide variety of forest stand conditions, ranging from single species even-aged to mixed species uneven-aged stands.

FVS provides a standardized growth model platform used across the US and parts of Canada. It currently contains 20 distinct regional variants, model equations specific to a geographic region. In addition, there are two Canadian variants and four hybrid variants, two that combine FVS with Oregon State University's ORGANON model, the Adirondack variant (Weiskittel et al. 2016) and the Acadian variant (Weiskittel et al. 2017). The northeastern US, including northern New England, falls within the suggested range of the FVS northeastern variant (FVS-NE).

Running FVS projections requires four components:

- FVS interface: For most users, this is the FVS graphic user interface (GUI) included with FVS. For some advanced users, the R programming and statistical language can be used to interact with FVS
- Input data: FVS requires stand and tree data. We discuss this in more depth in the "Preparing input data" section
- Variant code: In most variants, this is a shared library and we talk about this more in the Acadian model section
- Keyword file: The keyword record file tells the model what to do

Acadian model

Development of the Acadian model began in 2008 and has been continuously improved since then (e.g. Weiskittel et al. 2017). Recent model updates include revised height and diameter growth equations and a new approach to predicting mortality.

The Acadian model was developed to predict growth and mortality but not to simulate management activities or provide a user interface. We turned to FVS for those functions as the processes were already in place and many users are familiar with other FVS variants. Most FVS variants are written in the Fortran programming language (with some minor code in C and C++), and variants are compiled as DLLs (Dynamic Link Libraries) for use in Windows or shared libraries for use in Unix-based operating systems. The Acadian variant differs, using the FVS-NE variant DLL to handle the base FVS functions and R code to execute the Acadian growth and mortality calculations. This happens in the background, and the user experience differs slightly from other variants.

Getting started

The FVS software installation package is available on the FVS website https://www.fs.usda.gov/fvs/software/complete.php. The executable installer will install all of the needed FVS files. The default installation location is "C:\FVS". Additional details and installation instructions are available on the FVS website.

The FVS interface offers some functions to convert data into the SQLite database format that FVS uses and to view and edit the data. If you would like a more comprehensive set of features to view and edit input and output data, there are numerous graphic interfaces for SQLite. A few of the more popular open-source options are:

- SQLiteStudio: sqlitestudio.pl/
- DB Browser for SQLite: sqlitebrowser.org/
- DBeaver: dbeaver.io/

Preparing input data

FVS requires a stand table (FVS_StandInit) and a tree table (FVS_TreeInit). A description of the fields is available in the FVS user interface. The stand table contains information about the sampling design and the stand conditions. Refer to the Essential FVS user guide for more information about the format of the inventory sampling design. FVS-ACD uses a climate site index (CSI) instead of a traditional site index value based on ring counts from site trees. The CSI is a measure of potential height growth (meters) using climate and soil properties. CSI is available as a GIS raster layer. You will need to use your stand spatial data to generate a CSI value for each stand and place the CSI value in the FVS_StandInit table. That process is beyond the scope of this user's guide but should be relatively straightforward for someone with experience in GIS.

The tree data table is composed of records containing tree level information. Stand id, plot id tree id, species and diameter at breast height are the minimum required fields for each tree record. FVS species codes are available on the FVS User Guides webpage https://www.fs.usda.gov/fvs/documents/guides.shtml. FVS-ACD input tree data can include form and risk codes for hardwood tree records. To enter form and risk for a tree record enter code 54 in the Damage field and in the Severity field. Form is the first digit and risk is the second digit. For example, for a tree with form rating of F3 and risk rating of R2, the Severity code would be 32. Valid risk codes are 1 through 4 and valid form codes are 1 through 8. For additional information on hardwood form and risk rating see Pelletier et al (2016) and Castle et al (2017).

Starting FVS and setting up a project

Double click the FVS icon on your desktop to open FVS. The interface will start in your default web browser. The web browser simply provides the engine to power interaction with FVS and your data resides on your local computer. Nothing is transmitted over your internet connection.

At the top of the interface, you will see a title bar with information about the active project, the date and time you last accessed FVS, the release date, and the interface configuration.
Below the title bar you will see the main tab panel, consisting of six tabs: **Simulate**, **View Outputs**, **Visualize**, **View on Maps, Manage Projects**, and **Help**.

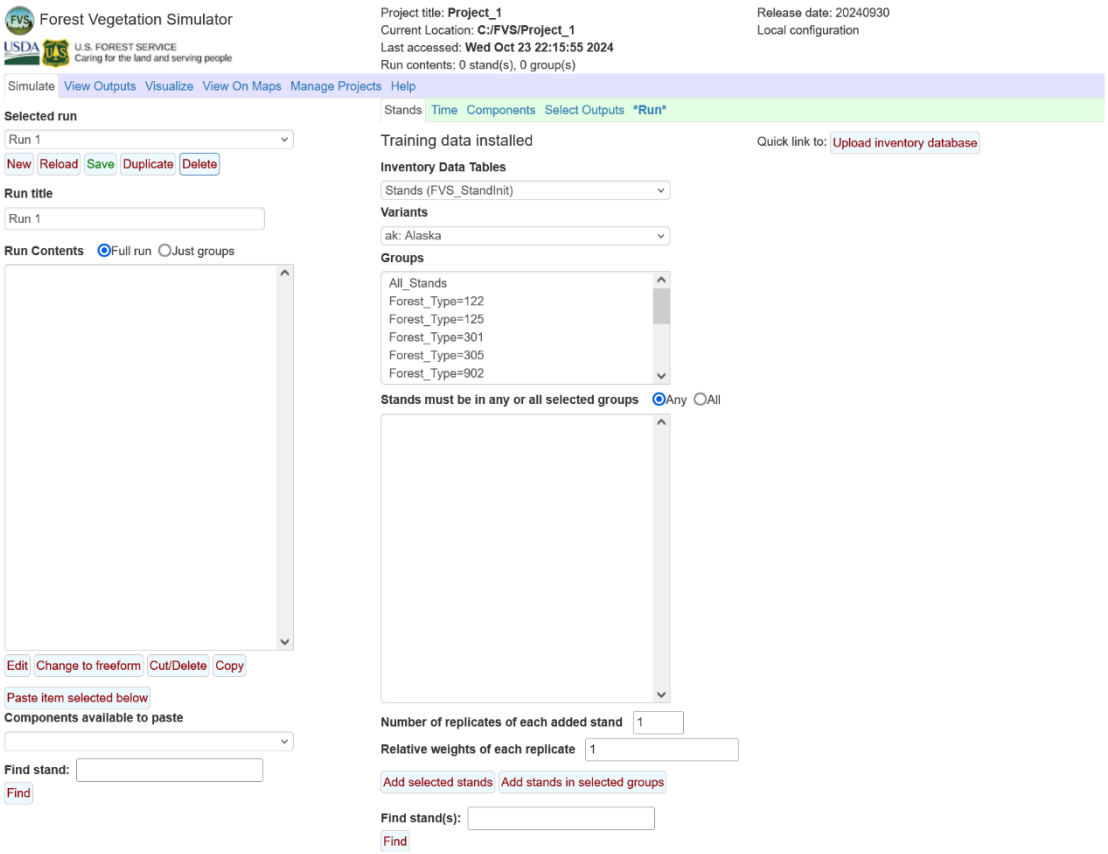


Figure 2. FVS graphic interface

When starting a new project in FVS, we need to create a new project folder. Each project folder will contain your input data, FVS keyword file and output data. This makes for easy organization of projects and a handy way to share projects. To create a new project, go to the **Manage Projects > Manage Project** tab. In the **New project title** field enter a descriptive title for your project and then click the **Make new project** button. Select the project you created and click the **Open selected project** button. This will cause FVS to open a new browser tab or window.

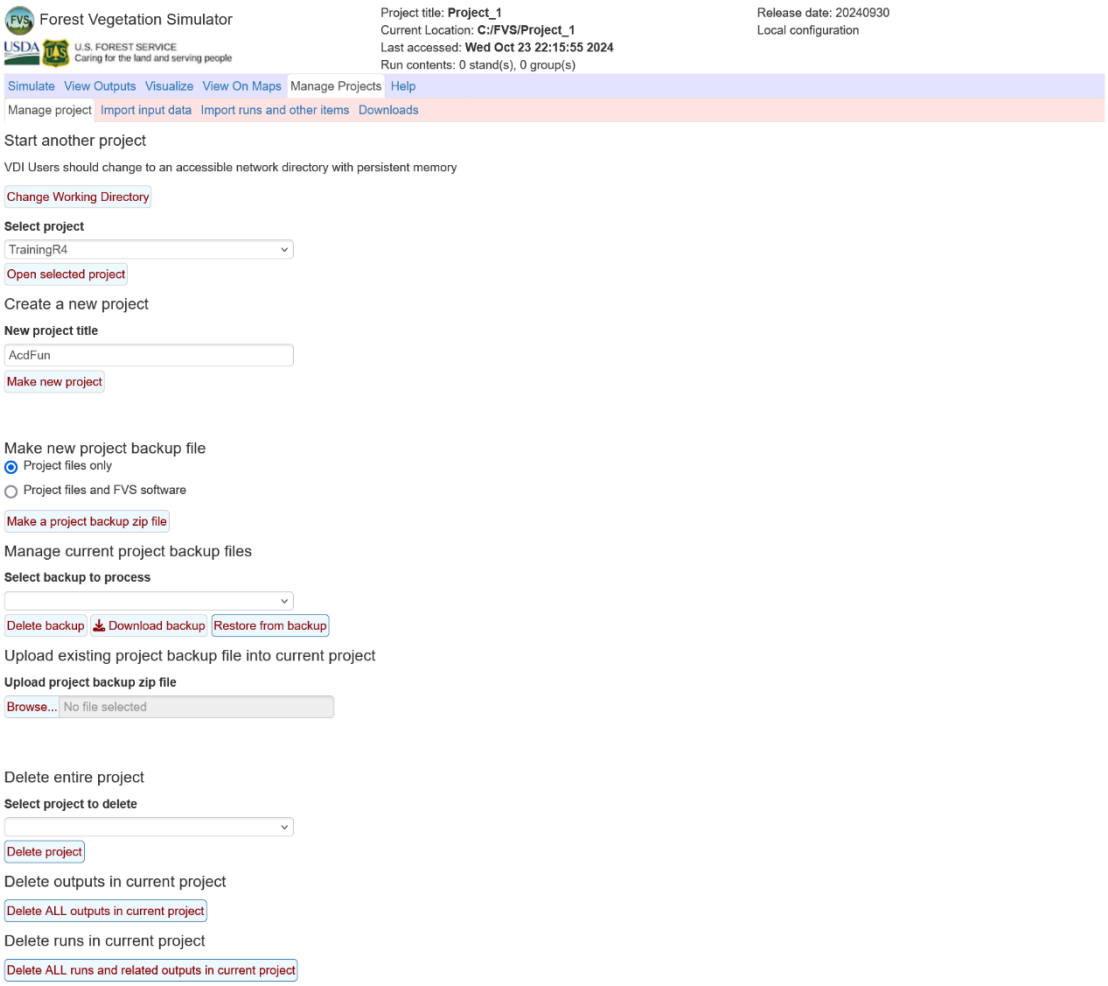


Figure 3. Manage project tab

Once your project is created you can create an input database. Click **Manage Projects > Import input data > Upload inventory database** tab. Under **Step 1: Upload FVS-Ready database** browse to inventory data in one of the following file formats: SQLite databases (.db or .sqlite), Microsoft Access databases (.mdb or .accdb), or Microsoft Excel spreadsheets (.xlsx). The file must contain a stand table and a tree list table with the field names matching the FVS data format. See the FVS website for a blank database, https://www.fs.usda.gov/fvs/software/data.shtml. Note that FVS field names are not case sensitive. Next under the heading **Step 2: Following upload** click the **Install uploaded database button**. This will create a new SQLite database, FVS_Data.db, in the project folder. Now you can view and edit your data.

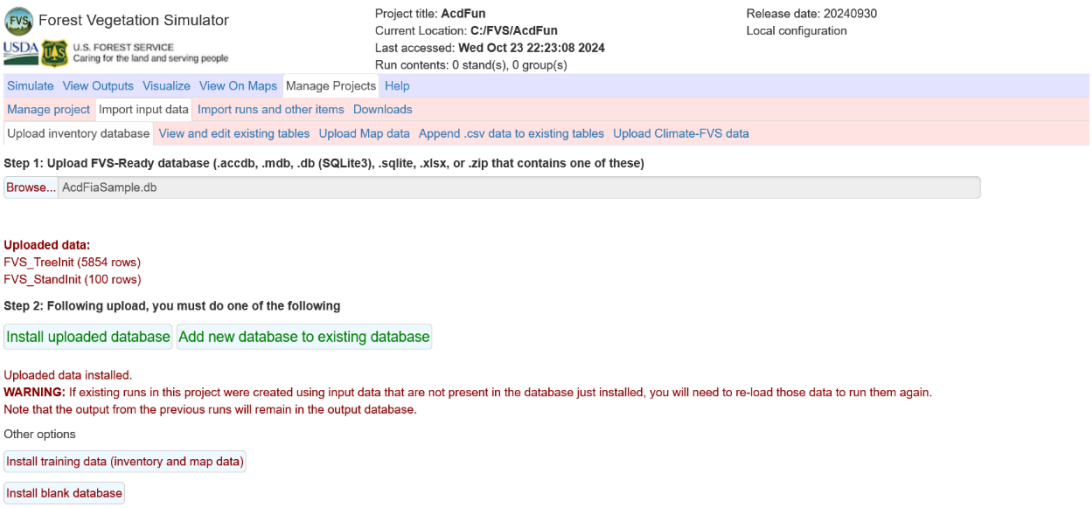


Figure 4. Upload inventory database tab

Click on **Manage Projects > Import input data > View and Edit existing tables**. The **Table to process** dropdown menu allows you to select a database table. The **Variables to consider box** allows you to choose which fields are displayed. Note that unselected fields aren't removed; they simply aren't visible. The **Select stands(s)** box and **the Find stand (ID)** text box will filter the table to the selected stands. A description of the table and its fields is located at the bottom of the screen.

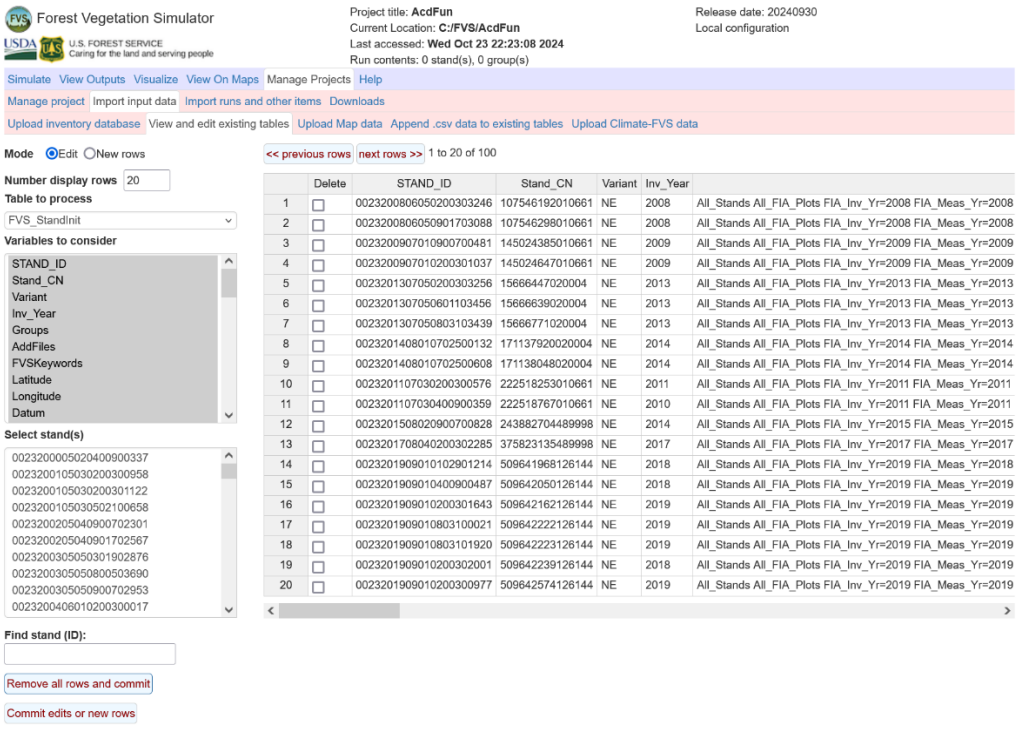


Figure 5. View and edit existing tables tab

Optionally you can also upload stand map data from a zip file, select the **Manage Projects > Import input data > Upload Map data** tab.

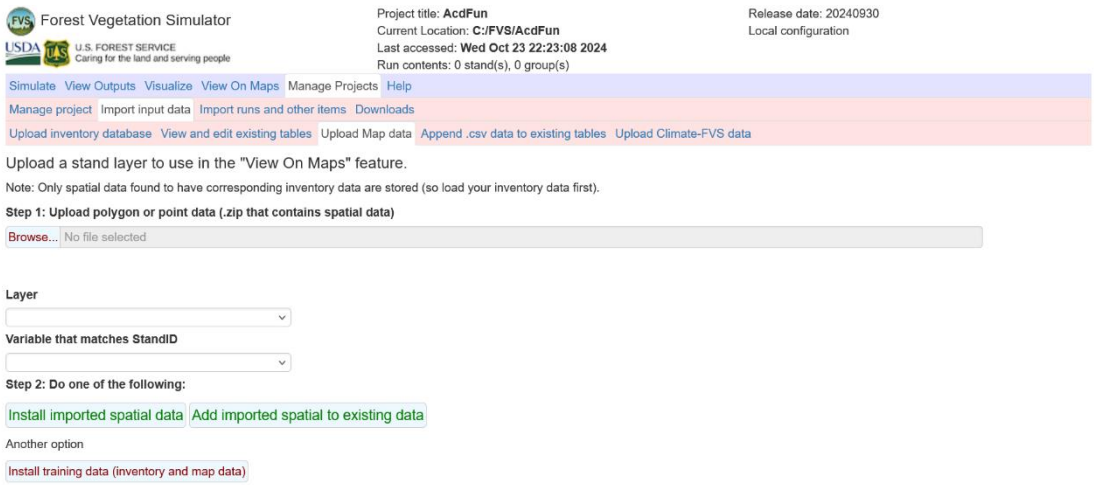


Figure 6. Upload map data tab

Basic projection

To begin a growth projection, click on the **Simulate > Stands** tab. Change the **Run title** in the text box to a descriptive name and click the **Save** button. If this is not the first run in your project, click the **New** button before assigning a new name in **Run Title**.

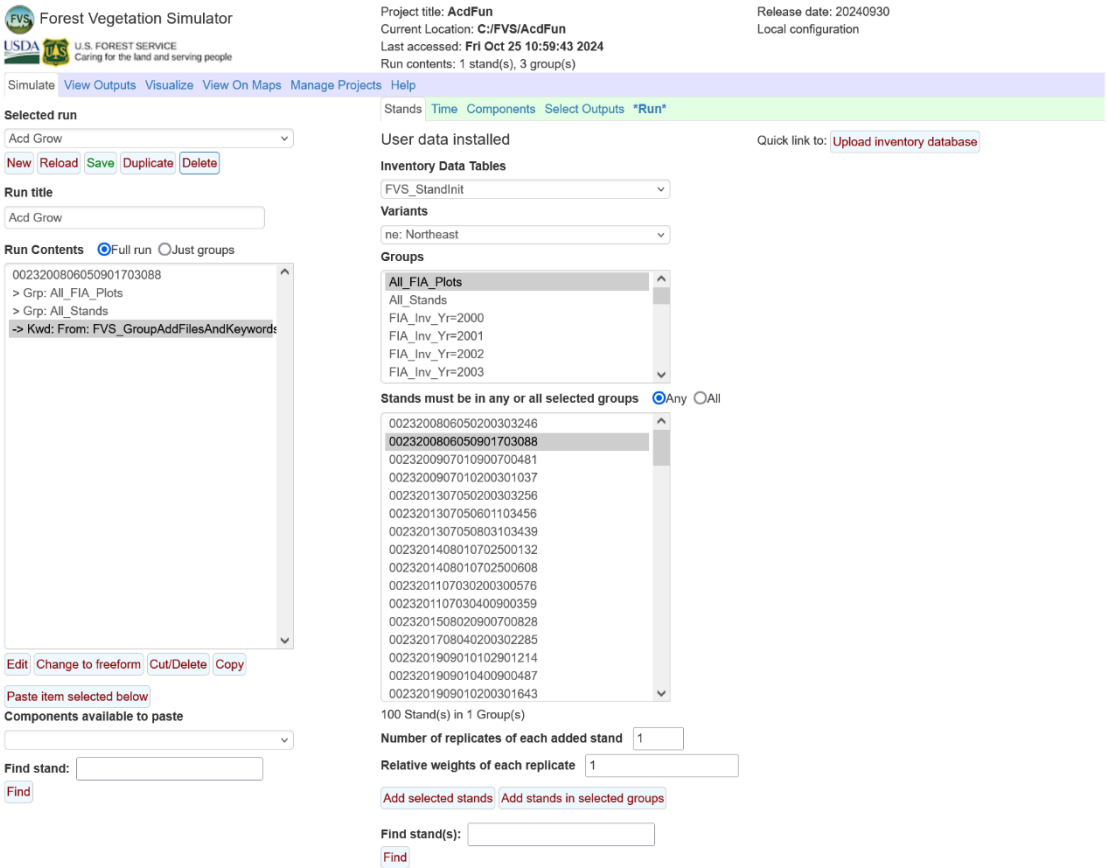


Figure 7. Stands tab

Stands can be selected to add to a projection by clicking on the Stand ID. The list of stands can be filtered with the **Variants** drop down list, the **Groups** list or the **Find stand(s)** search box. After selecting a stand or stands, add the stand to the projection by clicking **the Add selected stands** button. Once added to a run, the stand will be added to the **Run Contents** window

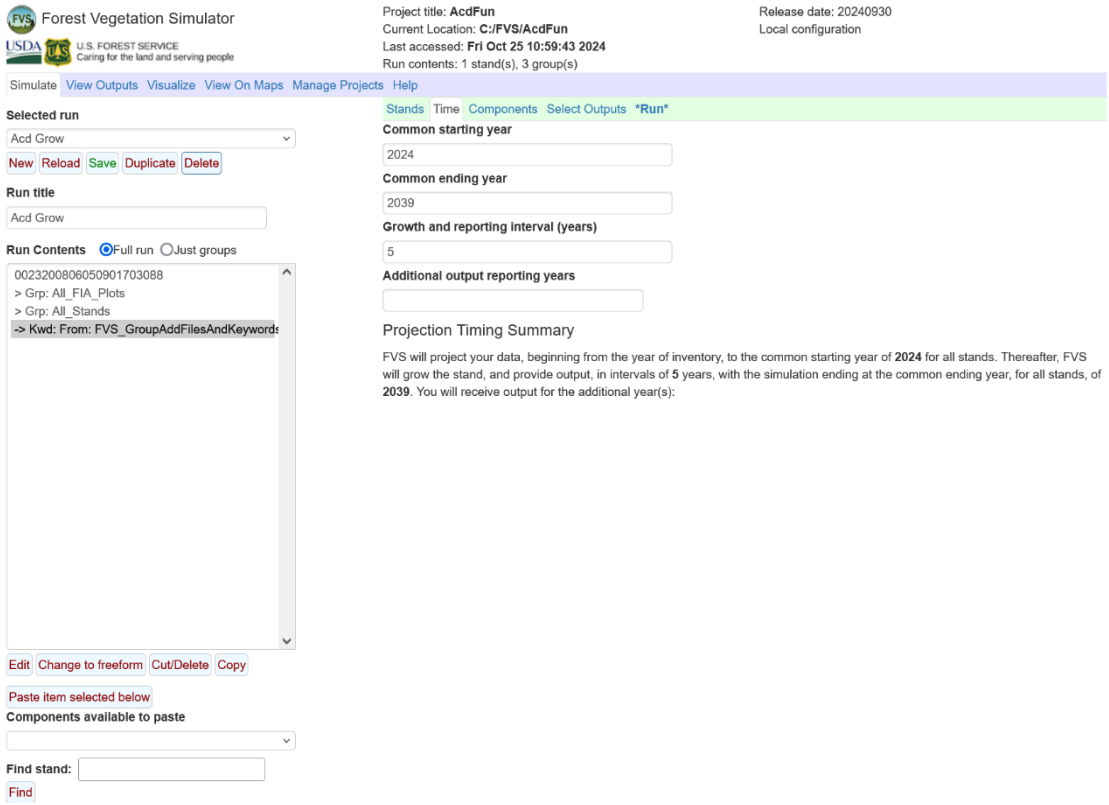


Figure 8. Time tab

Select the **Time** tab. The **Common starting year** field provides an option to project all stands to the same year to begin the projection. This is useful when you have inventory data from more than one year. The **Common ending year** is the last projection year. Unlike most other FVS variants the ACD variant projects growth and mortality in annual time steps. The **Growth and reporting interval (years)** field allows you to select the frequency that the ACD projections are written to the database. Keep in mind that FVS projections are limited to 40 reporting periods (cycles) per stand projected. The **Additional output reporting years** field allows FVS to write to the database in a year other than the selected reporting interval.

As noted above FVS-ACD uses CSI instead of a traditional site index value. Once we have our CSI value, we need to tell (base) FVS to select CSI from the database and pass it to FVS-ACD. This is accomplished by using a SQL query to select the CSI value for the current stand and then store the value in the event monitor. This can be done using the **Keywords** tab or since we already have the keywords written and stored in a (kcp) file, the **Editor** tab. Go to **Simulate > Components > Editor** and click the **Browse** button. Locate the AcdCsi.kcp file, select it and click **Open**. The contents of the kcp file will be shown in the editor window. Click the **Save in Run** button to add the code to the current run.

Select the **Select Outputs** tab and check the boxes for the FVS output tables you would like to have. The **Describe tables** drop down menu provides a brief description of the tables and details on each field in the table. It is always a good idea to select **Tree lists**. These tables will provide you with tree lists at the time of inventory and projected into the future, along with a list of harvested tree records and tree records immediately following a simulated harvest activity. By default, FVS (when accessed through the graphic interface) always produces FVS_Cases, FVS_Summary and FVS_Compute.

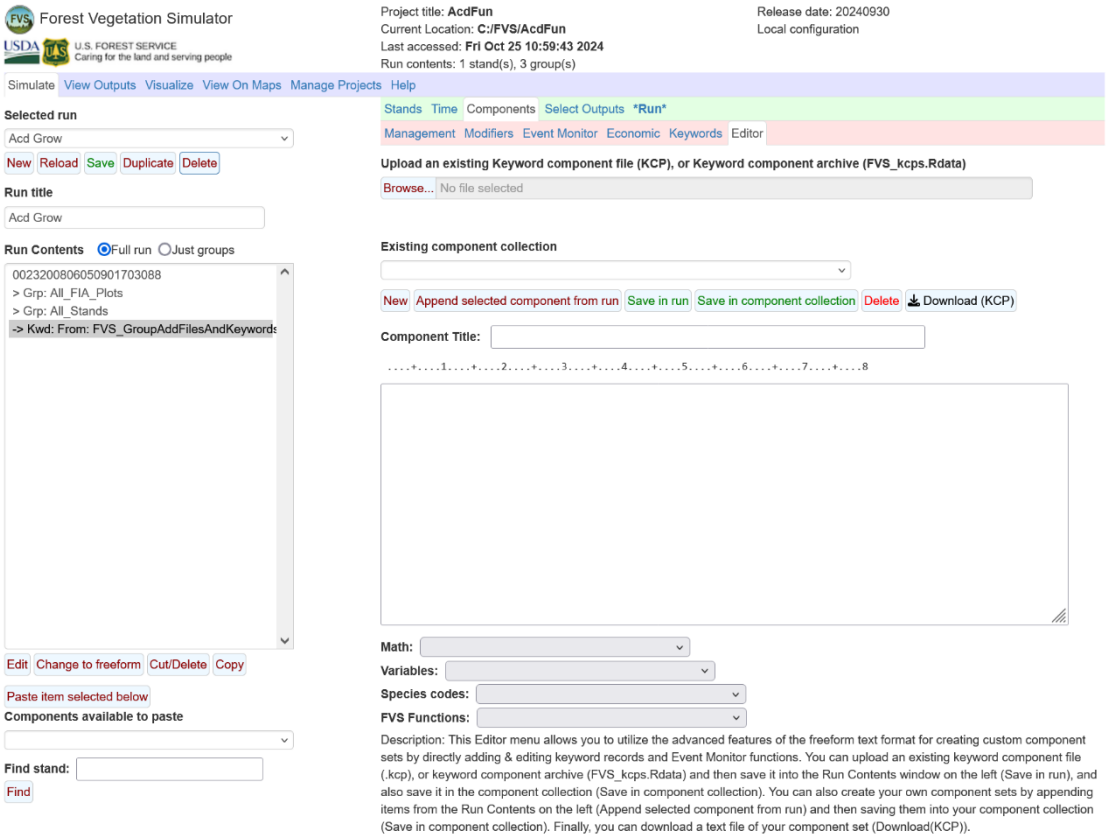


Figure 9. Editor tab

Select the ***Run*** tab. In the **MgmtID (4 chars)** text box create a management identifier up to 4 characters in length. This will be handy to track your projections if you create more than one projection scenario in the project.

In the **Select run script** drop down box, choose **AcadianGY**. This will make FVS-ACD options visible. These options are outside of the base FVS keyword architecture and are passed directly on to the FVS-ACD calculations. Following is a description of the options.

- **Simulate ingrowth**: Unlike FVS-NE, FVS-ACD can generate estimates of regeneration (full establishment model). Selection **Yes** will insert regeneration tree records into the tree list and selecting **No** will turn this feature off. The default is **No**.
- **Minimum DBH for ingrowth**: This is the minimum diameter threshold (cm) for ingrowth. The default is 3.0 cm.
- **Mortality model**: This option allows us to use the mortality estimates from FVS-ACD using the Acadian option or from FVS-NE using the Base Model option. The default is **Acadian**.

- **CutPoint**: The CutPoint option for the ingrowth model is the probability threshold where ingrowth will occur. The default is 0.95.
- **Merchantable volume logic**: The Acadian model can be used to calculate the total merchantable cubic feet (MCuFt) using the Li and Weiskittel regional taper equation by selecting the **Acadian** option. Selecting this option will not change the sawlog cubic feet (SCuFt) or board feet (SBdFt) calculations. The **Base Model** option will use FVS-NE to calculate all volume estimates. The default is **Acadian**.
- **Run with thinning modifiers**: Selecting the **Yes** option will use the post-thinning modifiers for balsam fir and red spruce for mortality, diameter growth, height growth and crown recession. The **No** option disables all post-thinning modifier calculations. **Yes** is the default option.
- **Run with Spruce Budworm modifiers**: The **Yes** option will use the spruce budworm defoliation modifiers for balsam fir and black, red and white spruce for mortality, diameter growth and height growth. The **No** option disables all spruce budworm modifier calculations. **No** is the default option.
- **Cumulative defoliation**: The cumulative spruce budworm defoliation is entered as a percentage. The default cumulative defoliation is 100.
- **Defoliation start year**: If using the spruce budworm modifiers, enter a year to start to defoliation. The default start year is 2020.
- **Defoliation duration (years)**: If using the spruce budworm modifiers, enter the number of years defoliation is predicted. The default duration is 10 years.

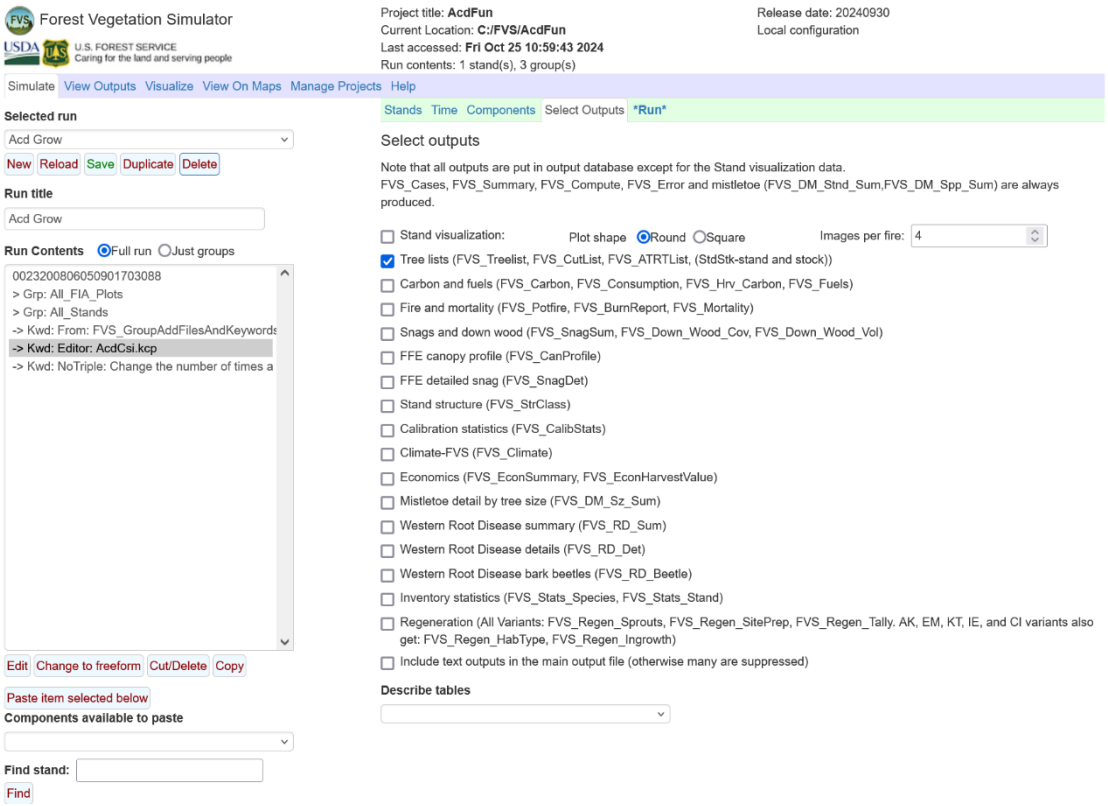


Figure 10. Stand outputs tab

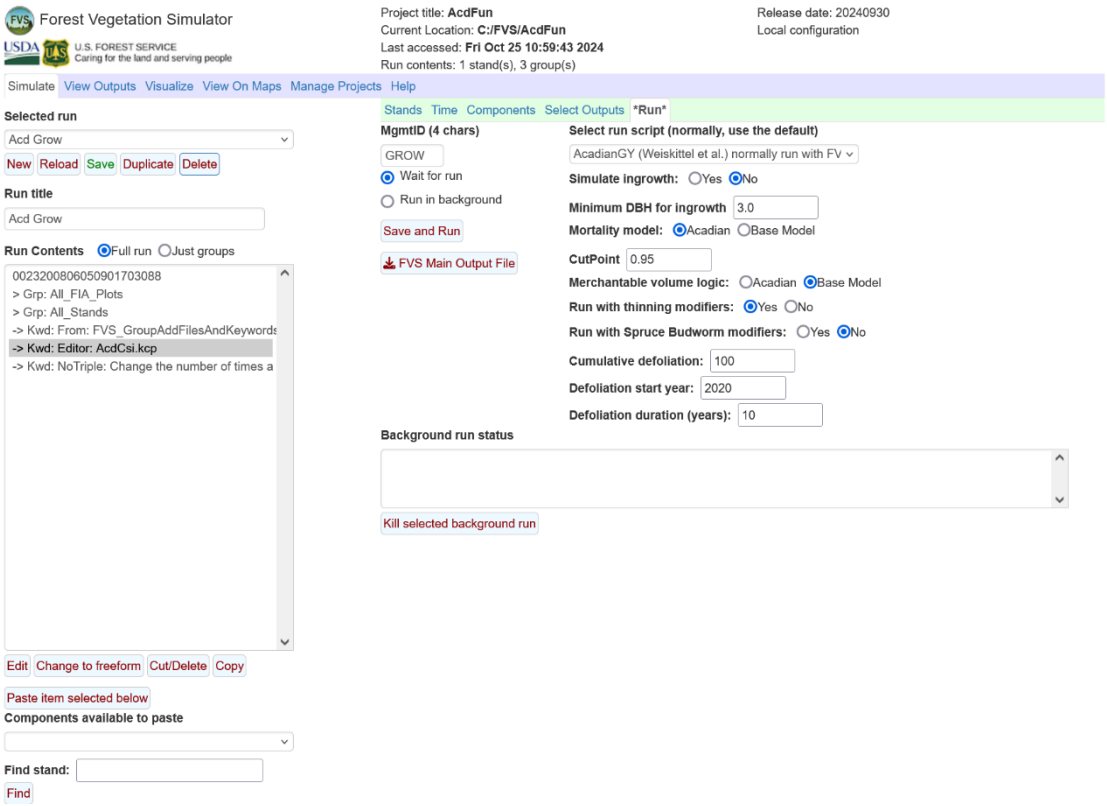


Figure 11. Run tab

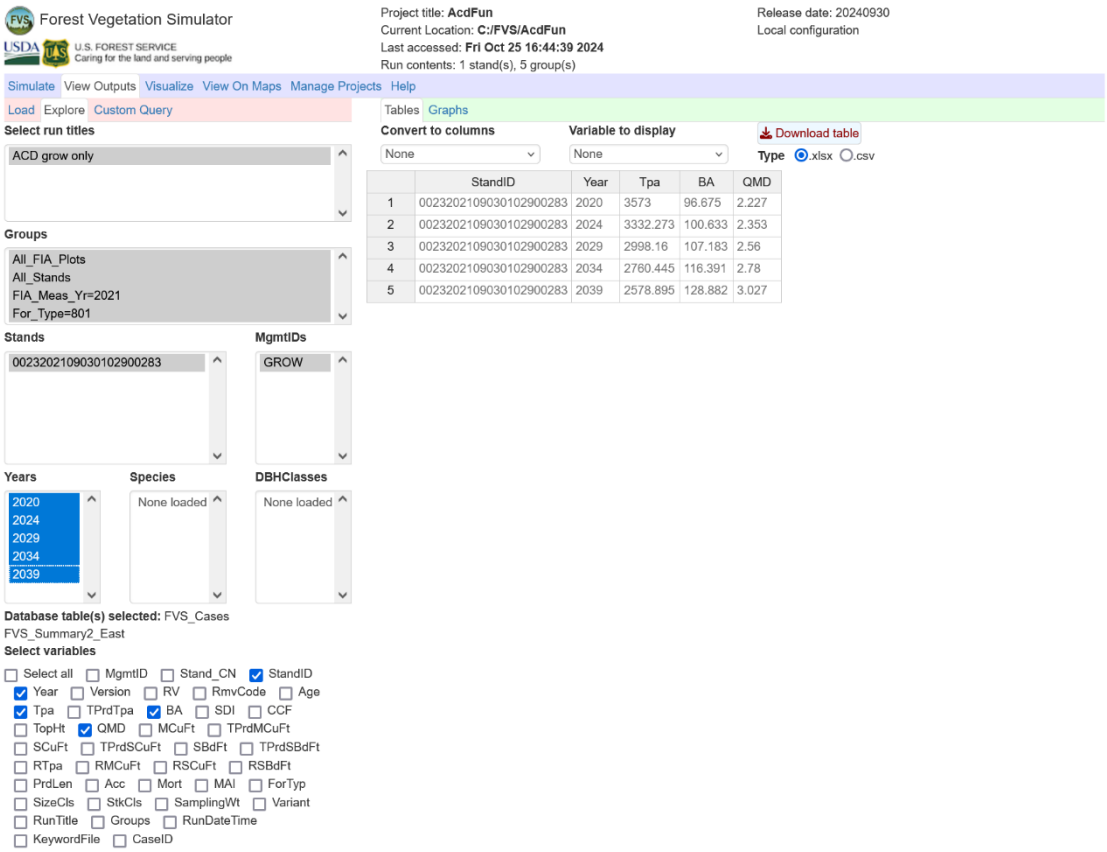
After selecting the FVS-ACD options, you can choose how FVS will run. By selecting the **Run in background** radio button, FVS will present the option to run projections on separate cores. This can speed up the processing time particularly for large projects. In our case with a single stand or a handful of stands, select the **Wait for run** radio button. Next click the **Save and Run** button.

Review results

The FVS user interface allows you to review your data and create tabular and graphical summaries in the **View Outputs** tab. Following are some basic workflows.

Create a simple summary table

Select the Load tab. In the **Runs to consider** box, select the run name that you created. In the **Database tables to consider box**, choose **FVS_Cases** and **FVS_Summary2_East**. You can review table descriptions and output field definitions using the **Describe tables** drop down menu. Select the **Explore** tab. You can filter the items displayed in the table through the selection boxes. In the **Stands** box click stand the stand you want to view. In the **Years** box select the years you would like to view. With the check boxes, select **StandID**, **Year**, **Tpa**, **BA** and **QMD**. Your table is ready to view and the **Download table** button will save your table to the location where your browser typically saves downloaded files.



	StandID	Year	Tpa	BA	QMD
1	00232021090301029002 83	2020	3573	96.675	2.227
2	00232021090301029002 83	2024	3332.273	100.633	2.353
3	00232021090301029002 83	2029	2998.16	107.183	2.56
4	00232021090301029002 83	2034	2760.445	116.391	2.78
5	00232021090301029002 83	2039	2578.895	128.882	3.027

Figure 12. View Outputs tab- summary table

Create a graph of basal area over time

Go to the **Graphs** tab. In the **Y-axis** box select **BA** and in the **X-axis** dropdown menu select **Year**. In the **Title** textbox enter “Basal area over time”. You can transfer your plot to a document by clicking the **Copy plot to clipboard** button and then pasting the figure into a document.

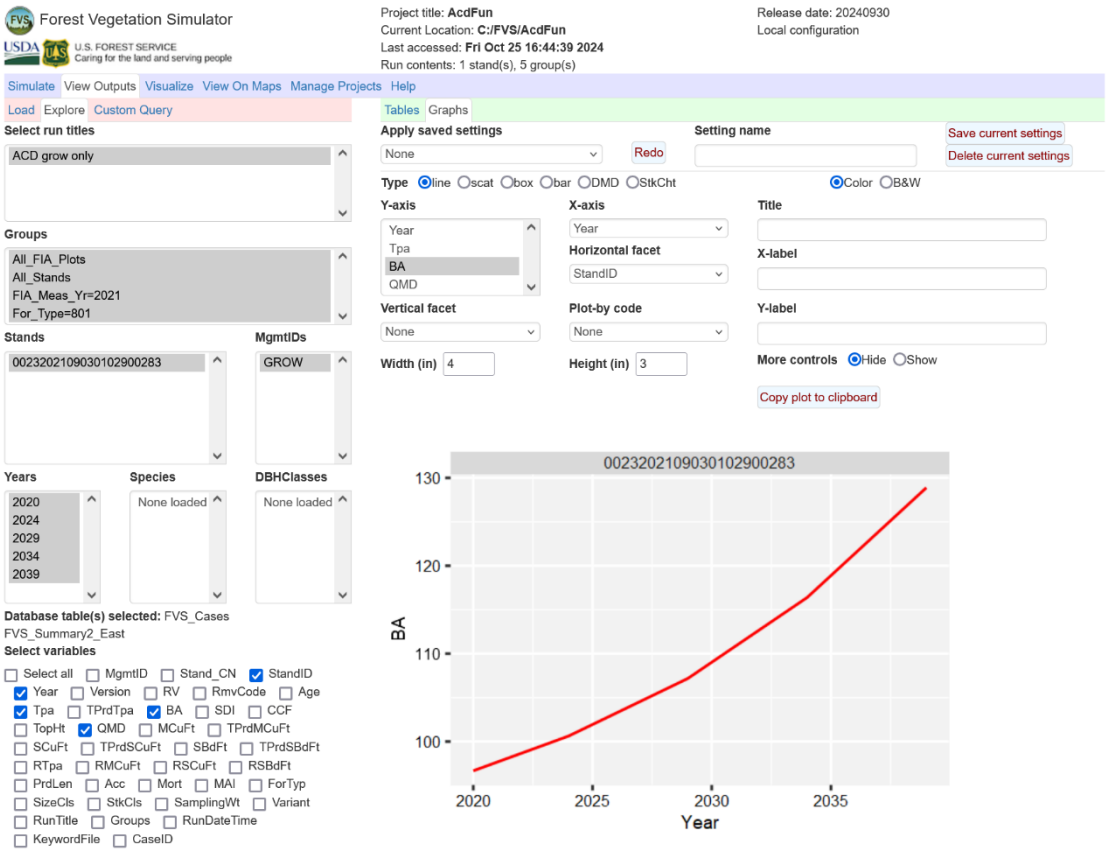


Figure 13. View Outputs tab- basal area over time figure

Create a graph of basal area by species and diameter class

Select the **Load** tab. In the **Runs to consider** selection box, select the run name that you created. In the **Database tables to consider** selection box, select **StdStk_East**. Go to the **Explore > Graph** tab. Select the **Years**, **Species** and **DBHClasses** you wish to view. Then select **Type – bar**, **Y-axis- LiveBA**, **X-axis- DBHClass**. If you selected more than one stand, MgmtID or year, you can use the **Vertical** and **Horizontal facet** dropdown menus to group the data into panels.

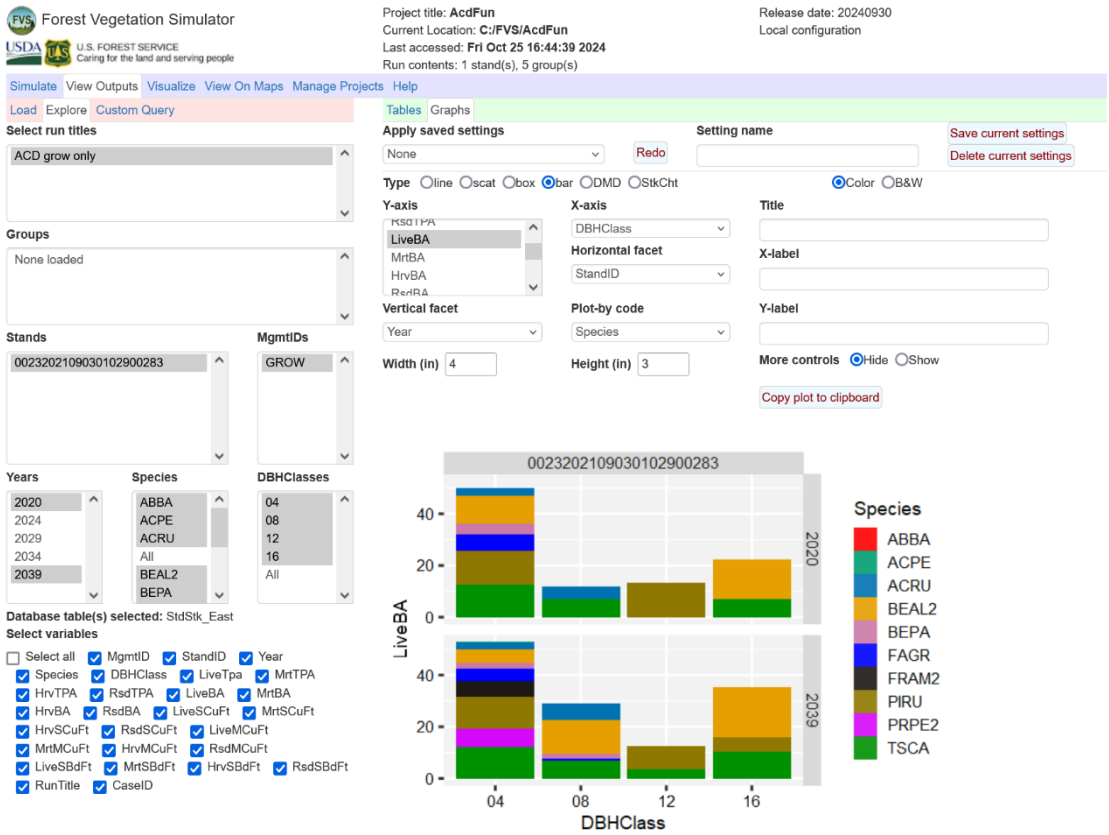


Figure 14. View Outputs tab- basal area species and diameter class

Thinning projection

Go to the **Simulate > Stands** tab. In this example we want to use the same stand as the previous projection. Click the **Duplicate** button. Change the **Run title** in the text box to a descriptive name and click the **Save** button. FVS has copied all of the information from the previous management scenario, so all we need to do is add a thinning. There are many options for thinning and the Essential FVS Guide is a good resource to learn more about the options.

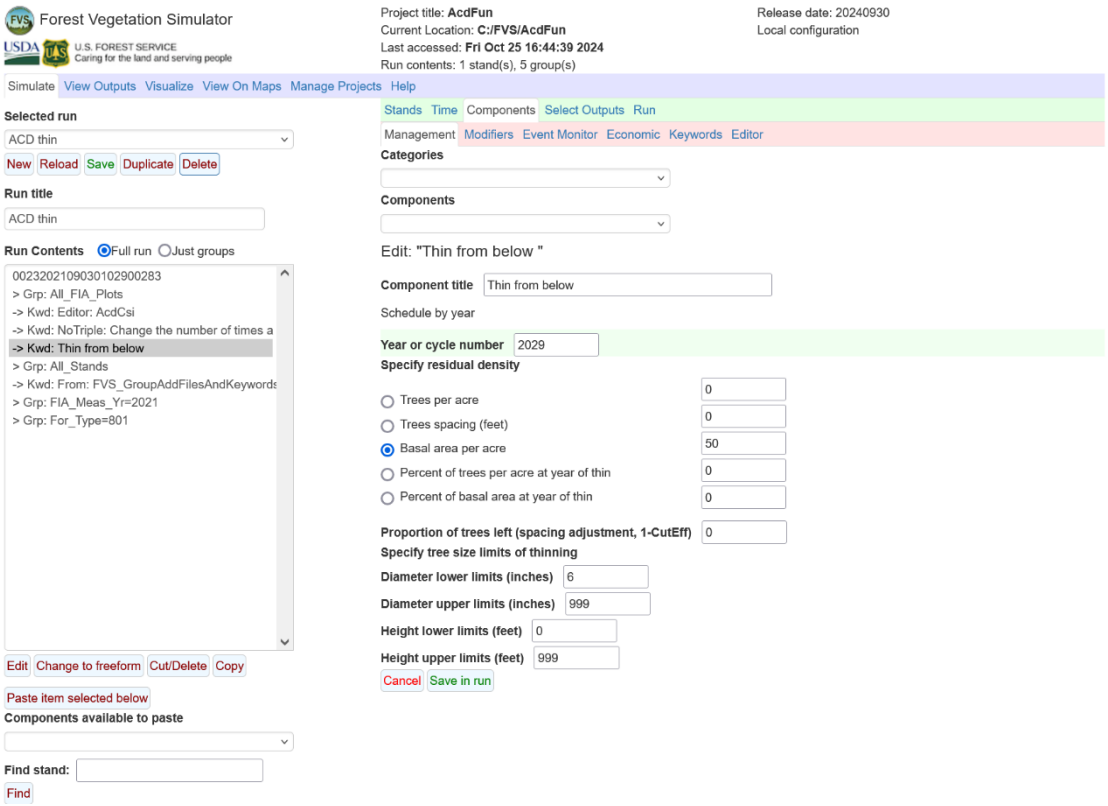


Figure 15. Management tab thinning scenario

In this example we will use the thin from below option. Go to the **Components** tab and then the **Management** tab. In the **Categories** dropdown box select **Thinning & Pruning Operations** and then in the **Components** dropdown box select **Thin from below**. Thinning activities can be scheduled using stand attributes such as basal area, height or relative density. Activities can also be scheduled by year. In this example we will use the year option. Click the **Schedule by year** radio button. Enter the desired schedule year. To specify the residual basal area, select the **Basal area per acre option** and enter the desired residual basal area value in the text box. Note that this basal area only includes trees between the **Diameter lower limits** and **Diameter upper limits**.

Select the ***Run*** tab. In the **MgmtID (4 chars)** text box create a management identifier up to 4 characters in length. In the **Select run script** drop down box, choose **AcadianGY**. Click the **Save and Run** button.

FVS-ACD options

Not all base FVS keywords have an effect on FVS-ACD projections. Most of the keywords affecting the model execution or simulated management activities should work as in any other variant. The differences are primarily related to keywords that affect growth or mortality predictions.

FVS-ACD contains maximum tree diameter and heights by species. These defaults can be overridden by using the TreeSZCp keyword. FVS calibration keywords can be used with FVS-ACD for diameter growth, height growth and mortality. BaiMult is used to set the species diameter growth multiplier. The HtgMult keyword is used to set the species height growth multiplier. The MortMult keyowrd is used to set the

species mortality multiplier within a diameter range (default: 0 to 999 inches). See the Keyword Reference Guide for the Forest Vegetation Simulator (Van Dyck and Smith-Mateja 2001) for more details on use of these keywords. The *Fix** family of multipliers (FixDg, FixHtg, FixMort) does not affect FVS-ACD calculations.

As noted in the *Basic projection* section, there are growth model option available outside of the FVS keyword structure. These include

- spruce budworm defoliation modifiers for balsam fir and black, red and white spruce for mortality, diameter growth and height growth
- post-thinning modifiers for balsam fir and red spruce for mortality, diameter growth, height growth and crown recession
- full ingrowth sub-model

By default, FVS-ACD also includes a hardwood form and risk modifier for mortality and diameter growth calculations for select hardwood species (northern red oak, sugar maple, yellow birch, red maple, paper birch, quaking aspen). If form and risk codes are present in the tree list data, these will be applied.

Learning more

FVS is a powerful program and mastery takes time and practice. A great place to beginning learning more is on the FVS website. The Forest Service offers live training sessions each spring. This is the best way to kickstart your experience with FVS. Another option is the online self-paced course, available on the FVS website
https://www.fs.usda.gov/fvs/training/selfpaced.shtml.

A wealth of FVS documentation is available at https://www.fs.usda.gov/fvs/documents/index.shtml. The following are essential to getting started with FVS-ACD

- Essential FVS: A User's Guide to the Forest Vegetation Simulator
- Northeast (NE) Variant Overview: Forest Vegetation Simulator
- Keyword Reference Guide for the Forest Vegetation Simulator

If you encounter a problem and are stuck, asking a colleague is probably the most fruitful avenue. We are all on this FVS journey together and someone in our sphere has probably had the same or a similar problem. The FVS help desk is available at https://www.fs.usda.gov/fvs/support/index.shtml. Note though that the helpdesk is not a substitute for learning FVS and does not provide support for the FVS-ACD equations.

If you think you have found a problem with base FVS, contact the FVS helpdesk at https://www.fs.usda.gov/fvs/support/index.shtml. For problems with FVS-ACD calculations contact Ben Rice (midgard.natural.resources@gmail.com) or Aaron Weiskittel (aaron.weiskittel@maine.edu).

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FVS-ACD quick start guide

Following is a list of the steps needed to establish a project, load data, set up a simple projection and run using FVS-ACD

1. New project

Go to the **Manage Projects > Manage Project** tab
Enter a title for your project in **New project title**
Click the **Make new project**

2. Input database

Go to **Manage Projects > Import input data > Upload inventory database** tab
Step 1: Upload FVS-Ready database browse to inventory file containing a stand table and a tree list table **Step 2: Following upload** click on the **Install uploaded database button**
This will create a new SQLite database, FVS_Data.db, in the project folder
To create optional stand map data, go to the **Manage Projects > Import input data > Upload Map data** tab

3. Growth projection

Go to the **Simulate** tab
Click the **New** button
Change the **Run title** in the text box to a descriptive name
Click the **Save** button
Select a stand or stands
Click the **Add selected stands** button
Select the **Time** tab
Enter a **Common starting year**
Enter a **Common ending year**
Enter a **Growth and reporting interval (years)**
Go to the **Keywords**> **Editor** tab
Browse to the CSI kcp file, open and click **Save in run**

4. Outputs

Go to the **Select Outputs** tab
Check the boxes for the FVS output tables you would like to have (**Tree lists** are recommended)
FVS always produces FVS_Cases, FVS_Summary and FVS_Compute.

5. Select run options and execute

Go to ***Run*** tab
In the **MgmtID (4 chars)** text box create a management identifier
In the **Select run script** drop down box, choose **AcadianGY**
Select FVS-ACD options
Select **Run in background** for large projects or the **Wait for run** for smaller projects
Click the **Save and Run** button.